

Regime maps for exact quantum compilation separate expressivity, certificates and feature transfer

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Quantum compilation is instance dependent, but a performance footprint alone does not distinguish an intrinsic compiler property from a proof limitation or a failed representation. We prove three model-specific results and organize them with a typed exact-regime record: a shared-tag Pauli block-encoding model has an all-size support-two ceiling; a rank-two dependent-triple model has intrinsic support number one and an explicit objective region where the proof applies; and a six-term linear-combination model has an exact pair-gain boundary. Weighted Clifford state preparation supplies an adverse transfer test. Its complete $n \leq 3$ domain contains 1,146 states, and a sign-aware 127-feature representation separates all labels into 1,109 cells. The representation is nearly injective rather than compact and does not transfer: parity-split lookup makes 32 errors on a frozen 120-state $n = 4$ panel and covers only two states. It shows no improvement under the specified 200-permutation shuffle null on this panel (mean 32.41 errors; empirical $p = 0.51$). An earlier prospective forecast also failed, matching 100 of 120 regime labels and 67 of 120 exact costs. The combined record rules out three otherwise tempting inferences: a proof ceiling need not be intrinsic support, certificate silence need not be necessity, and finite feature determination need not imply transfer. No universal compiler law or physical advantage is claimed.

Keywords: quantum compilation; exact optimization; stabilizer circuits; block encoding; algorithm selection; instance-space analysis.

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1 Introduction

Selecting an algorithm from instance features is a classical problem [1]. Modern instance-space analysis turns features and performance into interpretable regions that reveal where algorithms succeed or fail [2, 3]. The same perspective has reached quantum optimization and compilation. It has been used to study parameter initialization [4], quantum versus classical algorithm choice [5], compilation-option prediction [6], and device-specific compiler selection [7]. These studies establish that feature-conditioned performance mapping and algorithm selection are prior work, not contributions of this paper.

Exact compiler models permit a different question. When an exact optimum or a complete proof is available, one can ask why a reference compiler fails, whether a restricted family always contains an optimum, and whether the feature representation itself determines the label. These questions have different logical answers. A support bound may be intrinsic or merely sufficient for one proof. A proof can hold only for a region of objective weights. A feature map can fit a complete finite domain yet fail on new sizes. Combining these answers into one accuracy surface hides the distinction.

We therefore use a *regime map* as a typed reporting record rather than a single phase diagram. The record separates the reference-exact region, constructive improvement mechanisms, intrinsic support, proof-derived support ceilings, objective-conditioned proof validity, and feature determination with prospective transfer. A negative result is part of the record. For example, mixed feature cells prove that no classifier using those features can be exact on the stated domain, while a failed prospective forecast shows that an in-domain rule did not transfer. Missing coordinates are marked as unknown or not applicable rather than silently inferred from another coor-

dinate.

The evidence comes from four frozen quantum-compilation models (Table 1). Two Pauli block-encoding models admit all-size support results. A six-term linear-combination model admits an exact incumbent boundary. A weighted Clifford stabilizer-state model supplies complete finite graphs and a prospective size-transfer test. The models are deliberately heterogeneous. They are examples that separate logical coordinates, not a statistical sample of compilers.

The primary contributions are the three exact, model-specific results and the retained adverse transfer study. The regime map is their organizing schema. Together they establish a concrete comparison rule: numerical support, accuracy or region values may be compared only after their model, objective, domain and evidence class have been aligned. This changes the interpretation of several results. A safe syndrome-rank ceiling of five is not evidence that support five is intrinsic when a different proof gives intrinsic support one. Silence outside a proof-validity region is not evidence of necessity. Elimination of all finite-domain feature collisions is not evidence of transfer when the next-size test has almost no coverage. The distinctions therefore prevent specific overclaims that a single performance surface or separately reported headline numbers would leave open.

2 Regime-map definitions and study design

2.1 Exactness and support

Let $x \in \mathcal{X}$ be an instance, let θ specify objective weights, and let $\mathcal{C}(x)$ be the feasible compiler configurations. The exact cost is

$$C^*(x; \theta) = \min_{c \in \mathcal{C}(x)} C(c; x, \theta). \quad (1)$$

A deterministic reference compiler returns cost $C_D(x; \theta)$. Its exact region is

$$\mathcal{R}_D(\theta) = \{x : C_D(x; \theta) = C^*(x; \theta)\}. \quad (2)$$

Equality is always indexed by the model, domain and objective. A finite census, an all-size theorem and a frozen panel are reported as different evidence classes.

For models with a support coordinate $s(c)$, define the intrinsic support number by

$$\kappa(\theta) = \min\{k : \forall x \exists c^* \text{ with } s(c^*) \leq k\}. \quad (3)$$

A proof of sufficiency gives an upper bound on κ . Calling the bound intrinsic also requires tightness, such as infeasibility below it.

2.2 Constructive mechanisms and proof-validity regions

When the reference is inexact, a mechanism consists of a feasible witness, a strict cost reduction, and the structural edit that produces it. We use mechanism names only after these three parts are checked. A growing mechanism vocabulary does not weaken a separate theorem about the full feasible family.

A normalization proof may depend on θ . Its *proof-validity region* is the set of objective weights for which every local replacement is non-increasing and the analytic composition remains valid. Outside that set, the certificate is silent. A counterexample or a necessity theorem must be established separately.

2.3 Feature determination

Let $\phi(x)$ be a frozen feature vector and $y(x) = 1[x \in \mathcal{R}_D]$. A feature cell is the set of instances sharing one vector. A cell is mixed if it contains both labels. The minimum error attainable by any deterministic classifier using only ϕ on a finite domain is

$$E_{\min}(\phi) = \sum_v \min\{n_0(v), n_1(v)\}, \quad (4)$$

where $n_b(v)$ counts label b in cell v . Thus, $E_{\min} > 0$ is a representation-level impossibility result. Conversely, $E_{\min} = 0$ only proves determination on the registered domain. Cell count and singleton count reveal whether that determination is compressive or nearly memorizing.

2.4 Typed regime-map schema

For a declared model M , objective θ and domain $\mathcal{X}$, we record

$$\mathfrak{M}(M, \theta, \mathcal{X}) = (\mathcal{R}_D, \mathcal{W}, \kappa, B, \Omega, \phi, E_{\min}, \mathcal{T}; \alpha). \quad (5)$$

Here $\mathcal{W}$ is the set of verified constructive gap witnesses, B is a proof-derived support ceiling,

Table 1: The four compiler models answer different questions about exactness. “All sizes” means all sizes admitted by the stated algebraic model, not arbitrary compiler or hardware settings. Finite entries state the complete or frozen domain used.

Model	Exact authority	Headline result	Boundary that remains
Shared-tag Pauli block encoding	all sizes	exact optima need frame support at most 2	the smallest named support-two trade vocabulary is still refinable
Rank-two dependent-triple block encoding	all sizes	intrinsic support number $\kappa = 1$ under the unit objective	the support-one proof has an objective-dependent validity region
Six-term linear-combination compilation	all admitted batches; complete checks at $n = 1, 2$	the reference compiler is exact exactly at the proved pair-gain boundary	transfer beyond the frozen cost model is not claimed
Weighted Clifford stabilizer-state preparation	complete graphs for $n \leq 3$; frozen 120-state panel at $n = 4$	exact failures split into order, pivot, route and global mechanisms	an in-domain feature representation does not transfer to the frozen $n = 4$ panel

Table 2: Feature determination and transfer for weighted Clifford state preparation. The 127-feature representation determines the complete $n \leq 3$ labels but is nearly injective and fails the pre-specified transfer criterion.

Quantity	Complete $n \leq 3$	Frozen $n = 4$ panel
States	1,146	120
Feature cells	1,109	119
Singleton cells	1,072	118
Mixed cells	0	0 in-panel
Irreducible	0	0 in-panel
in-domain errors		
Parity-split lookup errors	–	32
Covered states in parity split	–	2
Shuffle-null mean errors	–	32.41
Empirical p (errors no larger)	–	0.51

Ω is the objective region in which its proof is valid, ϕ is a fixed feature representation, $\mathcal{T}$ is a prospective-transfer record, and α states the evidence class (all-size proof, complete finite census, frozen panel or bounded search). Each coordinate carries one of three statuses: *value*, *unknown* (defined but not established) or *not applicable* (the coordinate is absent from the model). An unknown value is never replaced by another coordinate; in particular, B does not become κ

without tightness, and $E_{\min} = 0$ does not populate $\mathcal{T}$. Two numerical coordinates are comparable only when their mathematical type, model family, objective and domain agree. Across different models we compare logical role and evidence class, not numerical magnitude.

2.5 Model register

We represent an n -qubit Pauli word, modulo global phase, by its binary symplectic vector. Its Hamming support is $\text{wt}(P)$ and $\langle P, Q \rangle \in \mathbb{F}_2$ denotes commutation parity. Full feasible sets and objectives are stated in Supplementary Section S1.

The shared-tag Pauli model contains three blocks, each with two nonidentity frames R_{j0}, R_{j1} satisfying $\langle R_{j0}, R_{j1} \rangle = 1$. A shared Pauli tag S gives the same two syndrome bits in every block, and those bits must differ. The objective charges frame, shared-tag and restored-target support, with one central frame per block weighted two and the other weighted four. The support coordinate is $\max_{j,k} \text{wt}(R_{jk})$.

The rank-two dependent-triple model contains two blocks. In block b , two anticommuting independent frames R_{b0}, R_{b1} determine $R_{b2} = R_{b0}R_{b1}$. Shared tags S_0, S_1 assign the same distinct nonzero two-bit labels to the two independent frames in both blocks. The objective weights central frame, noncentral frame, tag and

restoration support by $(t_c, t_{nc}, t_{tag}, t_r)$. The support coordinate is the maximum support of an independent generator.

The six-term linear-combination model takes six nonidentity Pauli words, with repetition allowed, and compares the six-singleton reference encoding with all 203 set partitions. Each block chooses whether to extract its common non-identity Pauli factor, and index ancillas may be shared or dedicated. The frozen objective is the sum of selection support, preparation-tree nodes and register width. Coefficients and amplitude-dependent preparation costs are outside the model.

The state-preparation model uses H , S , $S^\dagger$ and controlled-NOT gates with costs 1, 1, 1, 3. The reference is a deterministic Gaussian-elimination-style stabilizer synthesis. Exact costs are shortest-path distances from $|0 \cdots 0\rangle$ in the complete stabilizer-state graph. Complete graph sizes are 6, 60, 1,080 and 36,720 for $n = 1, 2, 3, 4$, respectively. Complete analyses use $n \leq 3$; the $n = 4$ study uses a frozen 120-state panel.

2.6 Verification and prospective discipline

All-size results combine written compositions with complete finite checks of the local obligations. Complete finite-domain results enumerate every admitted state or instance. Selected calculations were rerun from a clean environment, and separate verification code checked the stated theorem fields, finite-domain arithmetic and manuscript values without invoking the scientific calculations. These are same-team consistency checks, not external replication.

For the state-preparation transfer test, panel seed 20260821 generated 120 unique four-qubit states by applying 24 frozen random gates to $|0000\rangle$. Exact labels came from Dijkstra search over all 36,720 stabilizer states. The later representation concatenated 33 schedule/tensor coordinates, 53 reference-synthesis path coordinates and 41 sign-aware state coordinates. Index parity defined two folds. For each fold, a feature cell in the other fold predicted its majority label (ties negative); an unseen cell predicted negative. Coverage means exact feature-vector equality with a training-fold cell. The feature construction and criterion were fixed before this evaluation. The shuffle null permuted the 120 labels 200 times with Python seed 20260828, preserved

features and folds, and reran the identical lookup. Its empirical probability is the fraction of permutations with error count no larger than the observed count. Supplementary Sections S5–S6 give the complete coordinate definitions, deterministic reference compiler, freeze order and null algorithm.

3 Results

3.1 Shared-tag Pauli block encoding has a support-two ceiling

Theorem 1 (Support-two normalization). *For every size and target admitted by the shared-tag Pauli model under its unit raw-support objective, an exact optimum exists in which every frame Pauli has support at most two.*

The proof considers a frame R of support at least three, its anticommuting partner, and the shared tag. Each active qubit receives a two-bit class: its local commutation contribution with the partner and its local tag contribution. The partner relation makes the first class-bit parity odd. Every multiset of at least three such classes contains either a zero class or a repeated class. Removing the corresponding one or two letters preserves both parities, so feasibility is retained without modifying the tag.

The cost change is local. A complete 18,432-case check shows that restoration cost can never exceed the refunded frame cost. Repeated removal therefore reaches support at most two without increasing cost. A written parity argument covers arbitrary support; enumeration of 43,688 class tuples corroborates the support-two boundary and the support-three-and-higher descent.

The theorem closes the support envelope, not the smallest explanatory vocabulary inside it. Exact counterexamples successively require split anchors, support borrowing, off-support borrowing and hybrid constructions. These mechanisms remain within support two. Thus, intrinsic expressivity can be closed while a compact human-readable taxonomy remains refinable.

3.2 The rank-two model has intrinsic support one

Theorem 2 (Tight support-one normalization). *For every size admitted by the rank-two*

dependent-triple model under the unit objective, an exact optimum exists with generator support at most one. Support zero is infeasible, so $\kappa = 1$.

The decisive transformation localizes each dependent-triple block to one anticommuting core and then reconstructs the shared tag. Preserving the old tag during every local edit had produced looser bounds. Reconstruction exposes deletion credits that pay for core alignment and, when needed, relocation of the tag.

The local obligations are complete. They comprise 2,880 deletion cases, 6,912 core-alignment cases, 576 same-core tag-rigidity cases and 9,216 distinct-core tag lower-bound cases. The largest core-alignment penalty is three, while a deleted active column earns at least four. Distinct cores require tag cost eight; equal cores require cost four. These inequalities close every case in the all-size composition.

A conserved-syndrome abstraction gives a safe ceiling of five for this model. The tight theorem gives one. The difference shows that syndrome rank measures what a particular deletion proof preserves, not necessarily what an optimal compiler intrinsically needs. On the two models where both quantities and a local exchange margin are available, rank minus intrinsic support equals the measured margin. This is a two-point, rewrite-dependent observation, not a law.

3.3 The support-one proof has an objective region

Let $t_c, t_{nc}, t_{\text{tag}}, t_r$ weight commuting frames, non-commuting frames, tag support and restoration support. The same support-one proof remains valid when

$$2t_{nc} \geq 5t_r, \quad t_c + t_{nc} \geq 5t_r, \quad (6)$$

$$2t_{nc} \geq 2t_r + 2t_{\text{tag}}, \quad t_c + t_{nc} \geq 2t_r + 2t_{\text{tag}}. \quad (7)$$

When $t_c \leq t_{nc}$, the two inequalities involving $t_c + t_{nc}$ suffice for this certificate. In the displayed variable order, the unit structural objective is $(t_c, t_{nc}, t_{\text{tag}}, t_r) = (2, 4, 2, 1)$. Direct substitution gives margins $(3, 1, 2, 0)$ for the four displayed inequalities, so the point is inside and lies on the last boundary. The control $(3, 5, 2, 1)$ gives margins $(5, 3, 4, 2)$ and is strictly interior. Several outside controls violate at least one inequality. At any point outside, the certificate is silent. It does not say that support two is necessary.

A corrected search over 211,248 frozen-domain candidates found zero strict support-two witnesses under four objectives. This strengthens the domain-bounded negative result but does not prove global sharpness of the cone.

3.4 The six-term model has an exact structural boundary

Theorem 3 (Reference-exact boundary). *For every admitted six-term batch and size in the frozen linear-combination model, the reference construction is exact if and only if the proved pair-gain condition holds.*

For a subset B of the six terms, let $s(B)$ be the sum of their Pauli weights and let $f(B)$ count qubit positions at which every term in B has the same nonidentity Pauli. For a pair e , define $g(e) = 4f(e) - s(e)$. The exact condition is

$$\begin{aligned} g(e) &\leq 0, \\ g(e) + g(f) + 1 &\leq 0, \\ g(e) + g(f) + g(h) + 2 &\leq 0. \end{aligned} \quad (8)$$

The three lines quantify respectively over every pair e , every two disjoint pairs e, f , and every perfect matching $\{e, f, h\}$. The proof enumerates 203 partitions across 11 shapes and reduces every non-pair block to consequences of Eq. (8). Conversely, violation of each clause gives an explicit cheaper partition of shape $2 + 1 + 1 + 1 + 1$, $2 + 2 + 1 + 1$ or $2 + 2 + 2$. A separate complete regression checks 729 instances at $n = 1$ and 38,760 at $n = 2$ with zero mismatches. This compact boundary is a property of the declared model and objective, not evidence that other compiler boundaries are low order.

3.5 State preparation transfers mechanisms but not feature generalization

Exact shortest-path censuses on the complete $n \leq 3$ graphs identify 5, 28 and 189 reference-exact states out of 6, 60 and 1,080. Strict improvements fall into four mechanism classes. They arise from changing processing order, pivot choice, elimination route, or the global synthesis family. The last class shows that no tested row-reduction schedule closes the complete domain.

The prospectively frozen $n = 4$ forecast failed. It matched 100 of 120 regime labels and 67 of 120 exact costs. The mismatches were retained

rather than used to redesign the rule. This result rejects the proposed transfer criterion on the panel.

A coarse 13-feature representation provides a stronger in-domain negative. Twelve feature cells are mixed, which forces at least 43 errors among the 1,146 complete $n \leq 3$ states for every classifier using only those features. Adding sign-aware state structure changes the in-domain result. The 127-feature representation has no mixed cells and no irreducible in-domain error. It produces 1,109 distinct cells, including 1,072 singleton cells. This is finite-domain vocabulary existence, not a compact law.

The richer representation does not repair transfer (Table 2). On the frozen panel it forms 119 cells: 118 singleton cells and one pure doubleton cell. In the parity-split lookup test, only two states are covered and the remaining labels produce 32 of 120 errors. The specified 200-permutation shuffle null has mean 32.41 errors (range 30–33), and 51% of permutations have an error count no larger than 32. The rule therefore shows no improvement under this null on this panel; no equivalence or broader absence-of-signal claim follows.

4 Discussion

The four models support a conservative synthesis. Exactness regions are useful, but they do not have one authority or one explanation. A theorem about the full feasible family can coexist with an incomplete named taxonomy. A proof-derived search ceiling can be much larger than intrinsic support. An objective region can describe where a proof works without describing what is necessary outside. A feature vocabulary can determine a complete finite domain while carrying almost one cell per instance and failing at the next size.

The field-level consequence is a change in admissible comparison practice, not a universal performance prediction. A support ceiling may be entered as a search restriction but cannot be compared as intrinsic expressivity until tightness is shown. A proof-validity region may index a certificate but its complement cannot be labelled a necessity region. Zero mixed cells may establish finite determination but cannot be reported as generalization without a transfer coordinate. The present results supply a counterexample to

each prohibited inference. Recording the fields separately therefore changes which conclusions can be drawn from common exact-compilation outputs.

This separation clarifies the relation to algorithm selection. A performance footprint answers which method wins on which features. The present record asks additional questions when exact structure is available. Is there a constructive witness for the gap? Does every optimum admit a normal form? Is the normal form tight? Which objective weights preserve the proof? Does the representation determine the label? The answers can constrain or explain a footprint, but they do not replace instance-space analysis.

The state-preparation result gives the strongest warning against overcompression. Eliminating mixed cells by adding sign-aware features might appear to solve the boundary problem. The cell counts show otherwise. Most complete-domain states become singletons, and transfer coverage collapses to two panel states. The conversion is scientifically useful because it localizes the earlier failure to missing sign information. It does not produce a reusable predictor.

The same logic applies to support. The syndrome-rank ceiling remains useful for restricting search, even though it is not tight. The tight support-one result required a richer proof move, namely global tag reconstruction. Proof-language improvement, rather than larger enumeration alone, exposed the intrinsic value.

For practitioners, the reporting order is direct. State the exact model and objective. Record where the reference is exact. Attach constructive witnesses to failures. Separate intrinsic support from a safe proof ceiling. Index every certificate by objective weights. Test whether features determine the label before increasing classifier capacity. Freeze transfer tests when an exact referee is available. This order keeps favorable and adverse results on the same evidential footing.

5 Related work and contribution boundary

Algorithm selection and instance-space analysis already own the general idea of mapping feature-conditioned performance regions [1–3]. Quantum studies already select between quantum and clas-

sical algorithms [5], use instance spaces for quantum optimization [4], and predict compiler or device choices [6, 7]. We do not claim novelty for any of these tasks.

Exact, verified and algebraic circuit optimization is also established. Prior work supplies verified optimizer transformations [8], exact pattern matching [9], graph-theoretic simplification [10], and optimal Clifford synthesis on bounded qubit counts [11]. Benchmark suites characterize compiler behavior across circuits and devices [12, 13]. These works absorb any claim that exact optimization, formal checking, compiler benchmarking or Clifford synthesis is introduced here.

The Pauli models build on established unitary partitioning and Pauli compilation [14–17]. Connectivity overhead and routing constraints have also been characterized independently [18]. The present structural costs do not replace hardware-aware resource models.

The closest comparison for the map is instance-space analysis: both organize instance-conditioned outcomes, but an instance space does not by itself provide an all-size normal-form proof, distinguish a tight support number from a proof ceiling, or assign silence semantics to the complement of a certificate region. The closest comparisons for the model-specific results are exact and verified circuit optimization and bounded optimal Clifford synthesis. Those works establish optimizer correctness, exact local matching or bounded optimal circuits; they do not entail the two Pauli normalization theorems, the six-term condition in Eq. (8), or the present adverse feature-transfer result. Conversely, our results do not entail general verified optimization, practical pattern matching, hardware-aware compilation or optimal Clifford synthesis beyond the declared domains.

The primary advances claimed here are therefore the three model-specific exact results and the retained adverse transfer study. The typed map is an organizing and comparison schema that prevents their authorities from being conflated. The Supplementary Information gives the claim-by-claim subtraction by mathematical object, assumption, guarantee and consequence. The paper does not introduce a new optimizer, benchmark or algorithm-selection method, and it does not claim that the schema is a universal theory of compiler regimes.

6 Limitations

The all-size theorems concern two exact algebraic grammars and frozen structural objectives. They do not establish physical runtime, fault-tolerant resource savings, hardware advantage or production-scale compiler behavior. The six-term result is likewise restricted to its admitted batch family and cost decomposition.

The state-preparation conclusions are finite. Complete state graphs are used only through $n = 3$, while transfer is tested on one frozen 120-state $n = 4$ panel. The 127-feature representation is sign aware and permutation covariant, but it is near-injective in domain and has negligible parity-split coverage on the panel. Other representations, gate sets, costs or measurement-assisted preparation may behave differently.

The objective inequalities describe a sufficient region for one support-one proof. They are not proved necessary. A corrected frozen-domain search found no strict outside witness, but a witness in a broader domain remains possible.

The rank-minus-support observation uses only two measured models and changes under a different rewrite alignment. It should be treated as a diagnostic coincidence until a third model discriminates between competing explanations.

All computational checks were performed by the same research team. They check consistency and structural separation, but they are not independent external replication or external proof review. Public archival identity and third-party verification remain separate publication tasks.

7 Conclusion

Exact quantum-compilation studies can report more than a winner map, but the additional objects must remain distinct. Three model-specific exact results and one adverse transfer study show why: an intrinsic support result separates from a named mechanism vocabulary, a proof ceiling separates from a tight normal form, a certificate region separates from necessity, and in-domain feature determination separates from transfer.

The decisive adverse result is the state-preparation test. A rich feature map removes all mixed cells on the complete $n \leq 3$ domain but is nearly injective and fails on the frozen $n = 4$ panel. The appropriate conclusion is not that ev-

ery compiler has a simple regime geometry. It is that exact witnesses, proof scope, representation scope and prospective failures should be recorded separately before a compiler map is generalized.

8 Data and code availability

The review materials accompanying this version contain the exact result records, relevant analysis code, prospective protocols, per-state panel records, table-generation code and a separate verifier needed to check every headline number in this manuscript. The verifier checks theorem fields, finite-domain arithmetic, panel-partition invariants and manuscript values without invoking the scientific calculations. The materials also give a deterministic reproduction order and the declared Python dependencies. Detailed proof obligations and adverse-result tables are provided in the Supplementary Information.

No permanent public archive identifier or reuse licence is currently asserted.

Ethics, resources and author statement

This study used no human participants, animals, clinical records or personal data. Computation was used for exact enumeration, shortest-path optimization and finite proof obligations. Computational scale is a verification cost, not evidence of practical quantum advantage.

Sze Chun Yiu is the sole named author and retains responsibility for the scientific claims, analyses and final manuscript. Generative artificial-intelligence tools assisted with editorial refinement, literature-discovery support, code assistance and simulated review.

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